Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

A. follicle-stimulating hormone

B. oxytocin

C. pituitary gland

D. hypothalamus

E. gonadotropin-releasing hormone

F. luteinizing hormone

G. testosterone

Reasoning + Answer:

<think>

Firstly, let's recall the basic concept of a feedback loop in hormone regulation. The hypothalamus and pituitary gland work together in a feedback loop to regulate the production of sex hormones, such as testosterone in men. The hypothalamus secretes gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland to release follicle-stimulating hormone (FSH) and luteinizing hormone (LH). These hormones then stimulate the testes to produce testosterone. To maintain homeostasis, high levels of testosterone feed back to the pituitary gland to decrease the release of GnRH, FSH, and LH. Oxytocin is a hormone involved in social bonding and not directly in this specific feedback loop. Testosterone itself is a sex hormone produced by the testes, not a regulating agent. So, to identify the correct answer, we need to see if oxytocin, testosterone, or any other option is not part of the feedback loop or perhaps less directly involved compared